

Luis Antonio Goicouria, PhD

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LessWrong writeup · linkedin.com/in/luis-goicouria-phd · US work-authorized (no sponsorship required)

Scientist, developer, and evaluator of AI systems on scientific reasoning. As the first graduate student in a new epilepsy lab I built its recording platform from nothing and ran it for five years, learning where biological signal turns into artifact and believing a mechanism only once a perturbation could have falsified it. I then became a founding partner and lead developer at a software studio, shipping an Apple-featured visionOS editor and a live consumer health product. Since 2025 I have pointed the same discipline at AI: an interpretability study frontpaged on LessWrong whose main result is a warning about analysis — a representation that was perfectly decodable and, on ablation, causally inert — and evaluation tasks and rubrics that turn expert life-science judgment into criteria a grader can verify.

Experience

Tacit Labs — Life-Science Domain Expert (contract)

Jul 2026 – Present · Remote

- Design evaluation tasks and grading rubrics for frontier models at an AI × life-sciences lab (neuroscience, drug development): what is subtly wrong or unsupported by the evidence, turned into criteria a grader can verify exactly.

SkyPorch LLC — Lead Developer & Partner

Nov 2024 – Present · Oakland, CA

- Founding partner with end-to-end product and engineering ownership. Led **SpatialCut**, a native visionOS video editor **Apple featured on the front page of the Vision Pro App Store**, shipped within three months of my first line of Swift; full-stack on **Attached**, a live consumer health product (React Native, Swift, Python, Fly.io, CI/CD).
- Built and hardware-validated a real-time Bluetooth sensor-streaming iOS client for a wearable, with fault-injection testing (private work sample, on request).

Brown University — PhD Researcher, Laboratory of Epilepsy & Neurodevelopment

Jun 2018 – Oct 2023 · Advisor: Judy Liu, MD/PhD · Center for Translational Neuroscience

- **Built the lab's experimental capability from nothing** as its first graduate student — stereotaxic depth-electrode implantation, synchronized video-EEG/LFP acquisition, the pilocarpine and PTZ seizure models, histology and molecular readouts — and ran it for five years across four genetic models (PAR bZIP triple-knockout, CLOCK, ASH1L, DCLK1) and the drug model; the lab then ran on the methods I standardized.
- **Owned the data from surgery to figure:** months-long continuous within-subject recordings, acquired analysis-ready (synchronized video, event annotation, vigilance-state scoring); seizure scoring, power-spectral and excitability analysis on the acquisition toolchain (Pinnacle Sirenia, Clampfit, Prism), integrated with histology, molecular assays, and patch-clamp and field electrophysiology in the same animals; Python/R for behavioral data. Contributed the EEG acquisition and quantitative analysis behind the ASH1L preprint.
- **Reasoned from molecular cause to circuit phenotype to behavior:** bulk RNA-seq of hippocampus (DESeq2) followed to protein-level validation; dissertation on circadian transcription factors (CLOCK, PAR bZIP) and epilepsy. Mentored high-school through graduate researchers; 4 talks and 13 posters (AES, SfN, Gordon Research Conference); T32 predoctoral training grant.

Washington University in St. Louis — Undergraduate Research Assistant

Jan 2016 – May 2017 · Department of Surgery

- **TAMPITI randomized drug trial:** processed 100+ blood samples for pharmacokinetic and immunological analysis and screened 50+ Level-1 trauma patients on overnight shifts (co-author); first undergraduate selected for the Fuchs translational-immunology lab (platelet immune-modulation study; flow cytometry).

Selected Projects

Belief-State Geometry — what a model represents vs. what it relies on · [code](#) · [writeup](#) · 2025–2026

- Replication and original extension of Shai et al. (NeurIPS 2024), frontpaged on LessWrong; I own the experimental design, the analysis, and the writing, with AI-assisted implementation checked against tests and analytic ground truth. A small transformer trained on controlled hidden-Markov processes (PyTorch / TransformerLens); a linear probe recovers the analytic belief-state geometry from the residual stream ($R^2 = 0.989$ on Mess3). **The finding — *optimally but not minimally***: the model retains information that is provably useless for prediction and sheds it only under capacity pressure; ablating it collapses decodability to chance ($1.00 \rightarrow 0.50$) while leaving loss at the floor, whereas ablating what the model *does* use costs $\sim 20\times$ more — *encodes* without *causally driving*, shown by intervention, not fit. Ground truth by construction: 40 unit tests, pinned [uv](#), committed checkpoints, ~ 60 runs at 3 seeds per point.

arXiv Interpretability Radar · [code](#) — daily automated literature triage (RSS ingest, LLM relevance gate, deduplicating store, emailed digest), unattended on GitHub Actions since June 2026. **My-Local-SNP-Checker** · [code](#) — a local-first Python tool that annotates a consumer-genetics export against **ClinVar (NCBI)** on-device.

Skills

Evaluation & model characterization. Evaluation-task and rubric design for frontier models on expert scientific reasoning · LLM-as-judge scoring in production · agentic evaluation harnesses (Inspect, Docker sandboxes) · transformers and recurrent networks trained from scratch (PyTorch / TransformerLens); linear probing and causal ablation with positive controls · experimental design, controls, and statistics applied to model behavior.

Experimental neuroscience. Chronic depth-electrode video-EEG/LFP in mice (built from scratch) · stereotaxic surgery · pilocarpine and PTZ models; genetic lines, colony management, genotyping and primer design · whole-cell patch clamp; fEPSP and population-spike input–output · immunohistochemistry, Fluoro-Jade B / Nissl, confocal imaging, Fiji/ImageJ quantification · Western blot, PCR, bulk RNA-seq (DESeq2) · primary neuronal culture · flow cytometry (undergraduate).

Software. Python (NumPy, SciPy, pandas, PyTorch) · R · Swift / SwiftUI / visionOS · React Native / TypeScript · production systems with CI/CD (Fly.io) · GitHub Actions pipelines · reproducible experiment code (pinned [uv](#), pytest) · AI-assisted engineering (Claude Code) with tests as the check.

Publications & Writing

- **Goicouria L.** *Toy transformers may represent belief-state geometry optimally but not minimally.* LessWrong, June 2026 — frontpaged technical writeup with open code, checkpoints, and tests.
- Papendorp CM, Nolan E, Higashimori H, Jhanji M, **Goicouria L**, et al. (Liu JS-H, senior author). *Mutations in ASH1L cause a neurodevelopmental disorder with sex differences in epilepsy and autism.* **bioRxiv** 2025. [doi:10.1101/2025.02.21.639570](https://doi.org/10.1101/2025.02.21.639570) (preprint) — EEG acquisition and power-spectral analysis.
- **Goicouria L**, Liu J. *E-I E-I Woe: Mossy Cell Regulation of Granule Cell Activity in Temporal Lobe Epilepsy.* *Epilepsy Curr.* 2020;20(3):173–174 — first-author commentary.
- Zhang K, ... **Goicouria LA**, ... Jez JM. *Investigating the reaction and substrate preference of indole-3-acetaldehyde dehydrogenase from Pseudomonas syringae.* *Biosci Rep.* 2020;40(12).
- Spinella P, ... TAMPITI Investigators (incl. **Goicouria L**). *The immunologic effect of early IV tranexamic acid in severe traumatic bleeding (TAMPITI): a randomized trial.* *Front Immunol.* 2020;11:2085.

Education

Brown University — PhD, Neuroscience · 4.00/4.00 · Oct 2023 (*Circadian-Rhythm Protein Dysfunction and Epilepsy*). **Washington University in St. Louis** — BA, Biochemistry & Philosophy-Neuroscience-Psychology · May 2017.

Beyond work: tabletop role-playing modules and supplements as Whitewater Legend Lore. English (native), Spanish (proficient). Updated September 2026.